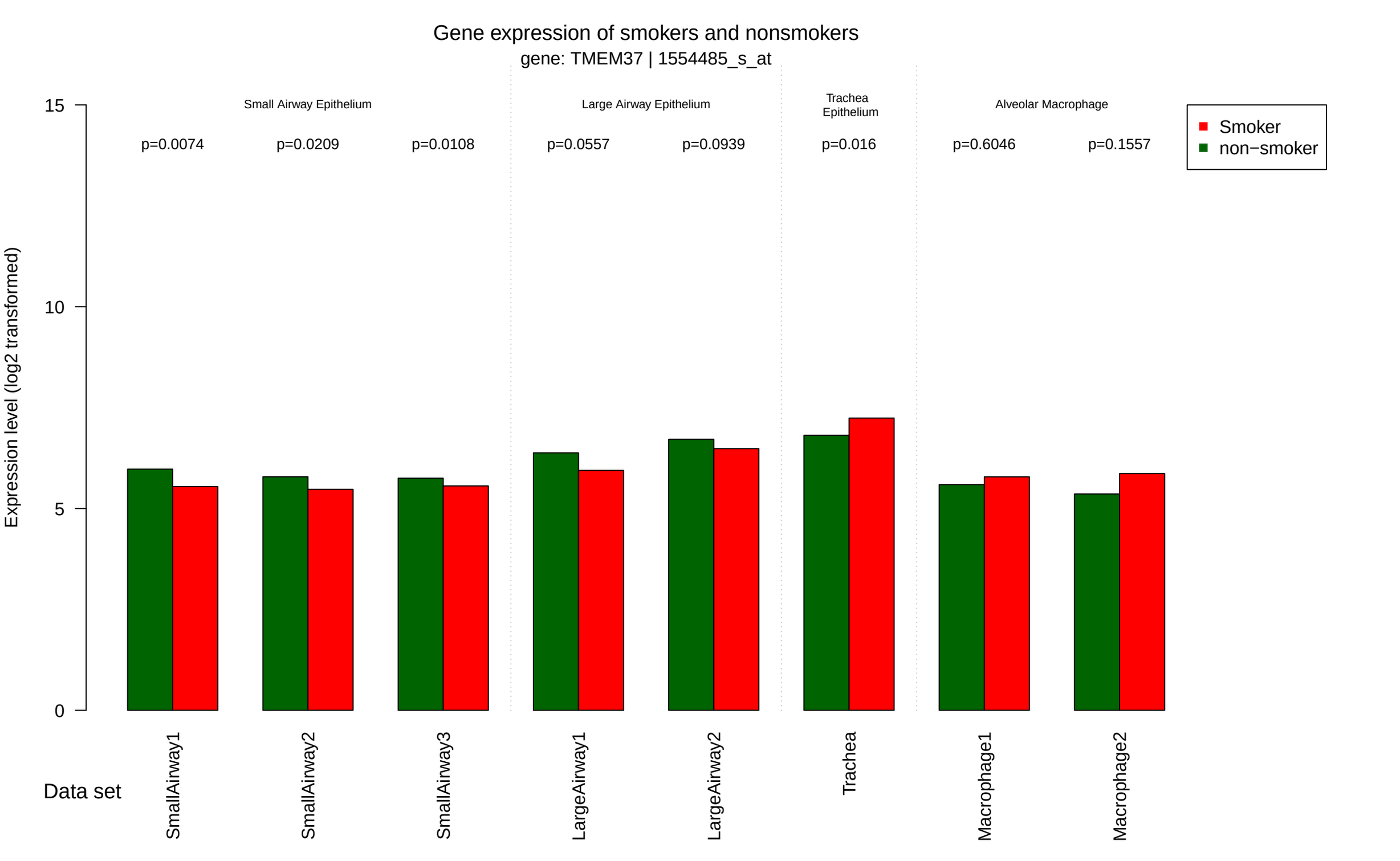
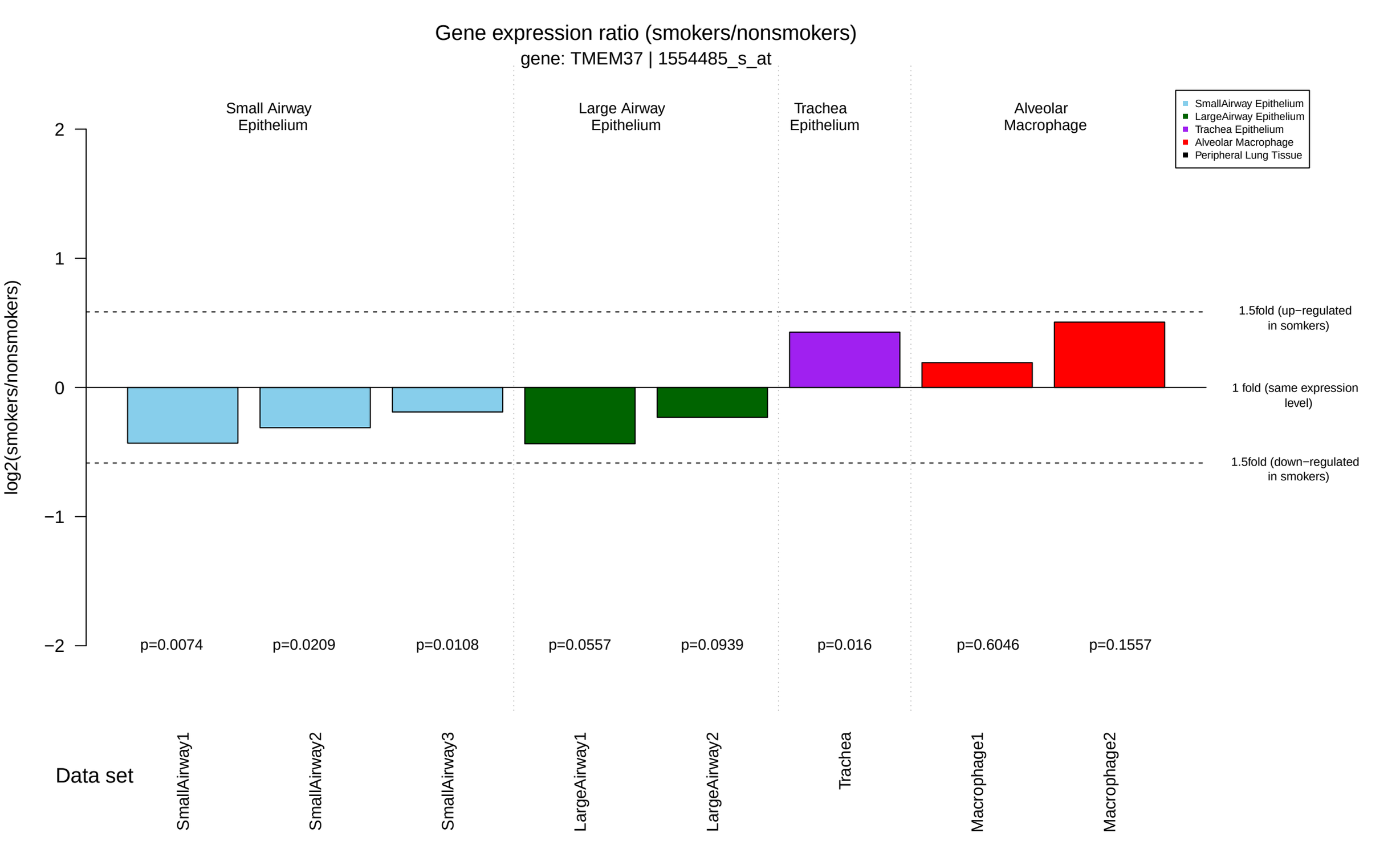
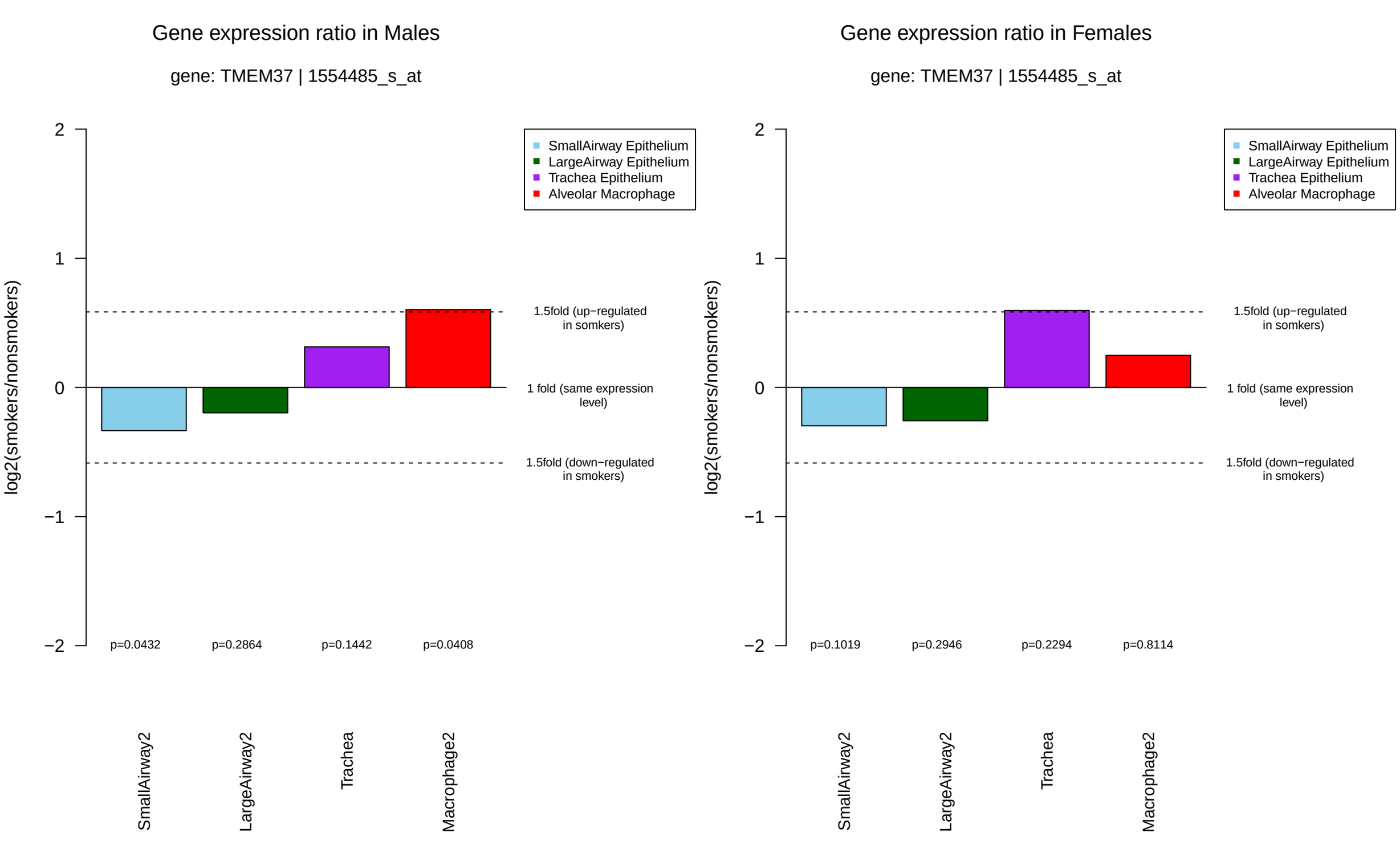
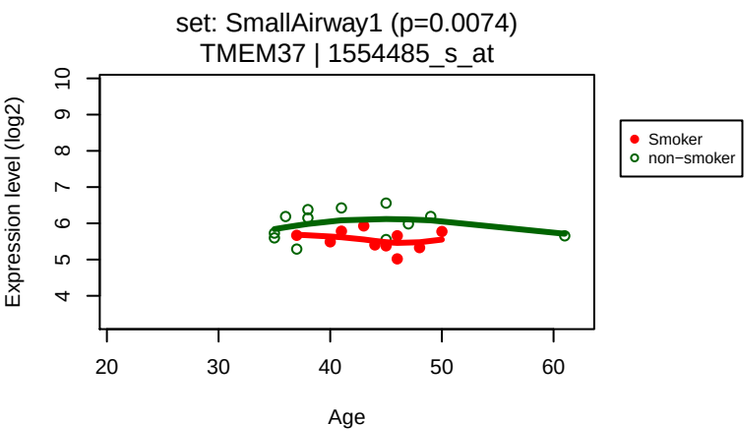
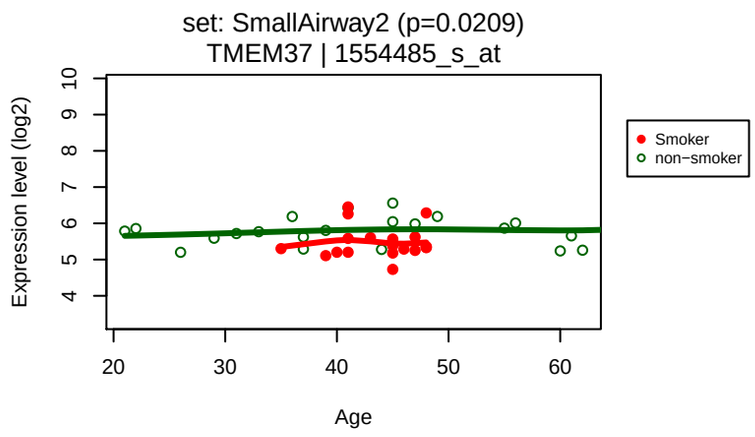
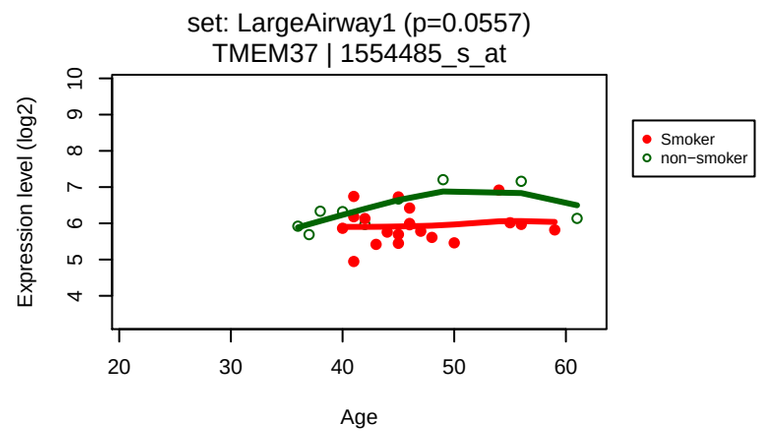

Gene expression of smokers and nonsmokers
gene: TMEM37 | 1554485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0074
p=0.0209
p=0.0108
p=0.0557
p=0.0939
p=0.016
p=0.6046
p=0.1557
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

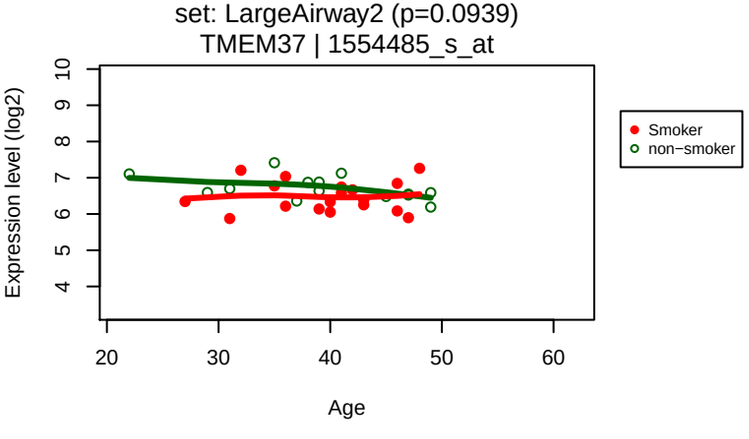

Gene expression ratio (smokers/nonsmokers)
gene: TMEM37 | 1554485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0074
p=0.0209
p=0.0108
p=0.0557
p=0.0939
p=0.016
p=0.6046
p=0.1557
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

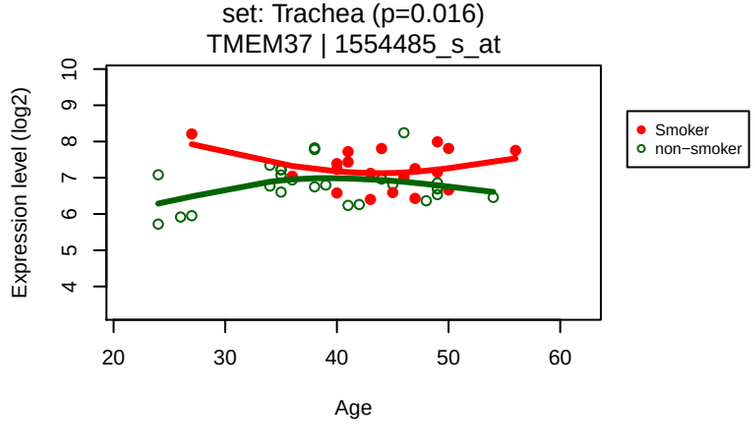

Gene expression ratio in Males
gene: TMEM37 | 1554485_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0432
p=0.2864
p=0.1442
p=0.0408
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM37 | 1554485_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1019
p=0.2946
p=0.2294
p=0.8114
SmallAirway2
LargeAirway2
Trachea
Macrophage2

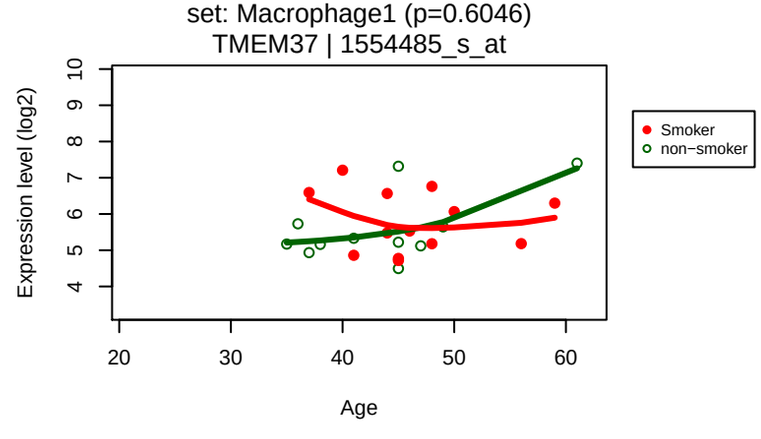

set: SmallAirway1 (p=0.0074)
TMEM37 | 1554485_s_at
Expression level (log2)
Age
Smoker
non−smoker

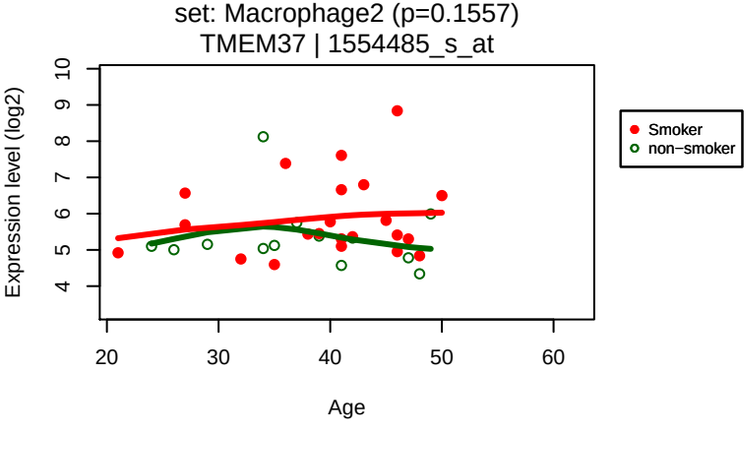

set: SmallAirway2 (p=0.0209)
TMEM37 | 1554485_s_at
Expression level (log2)
Age
Smoker
non−smoker

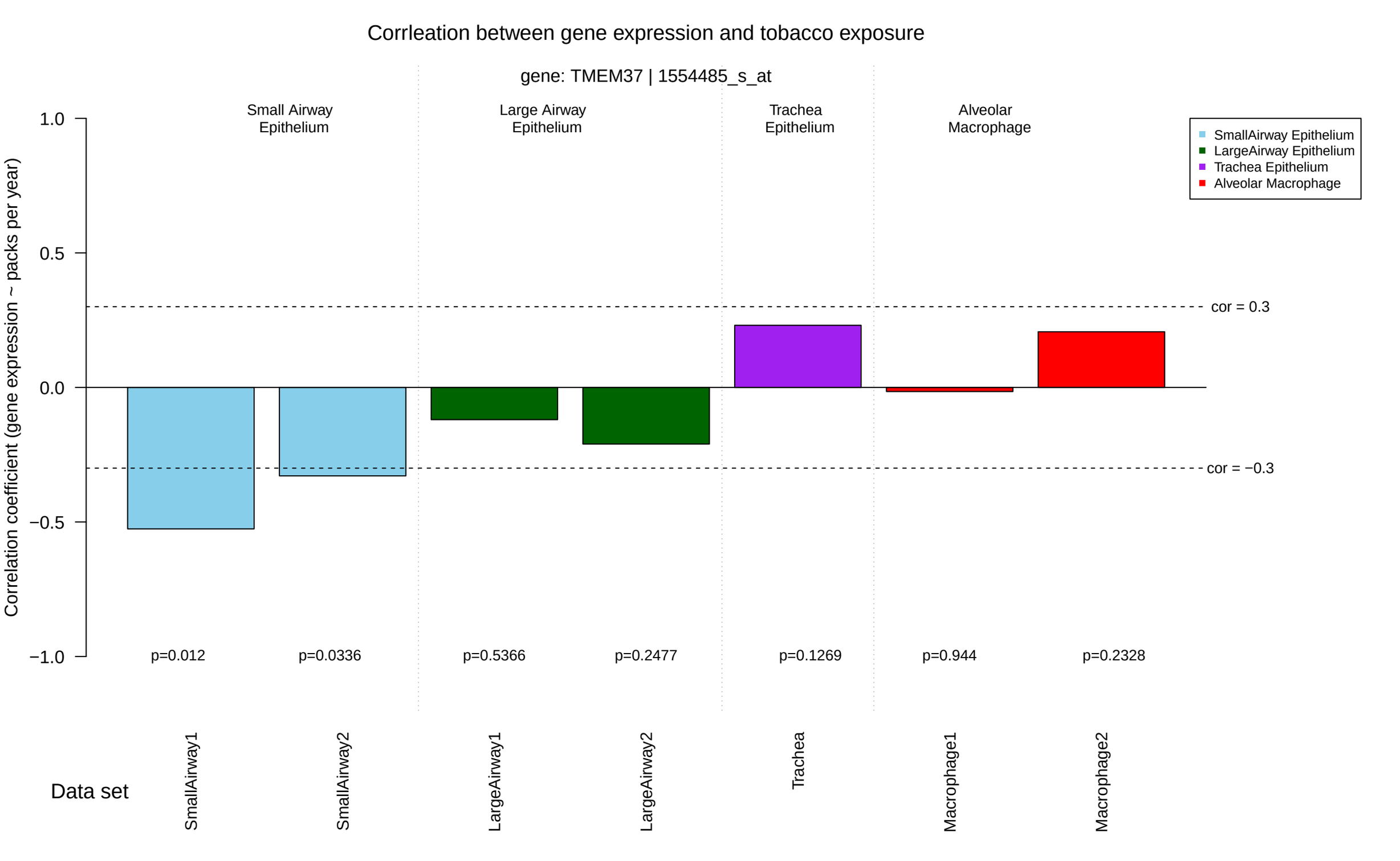

set: LargeAirway1 (p=0.0557)
TMEM37 | 1554485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0939)
TMEM37 | 1554485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.016)
TMEM37 | 1554485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6046)
TMEM37 | 1554485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1557)
TMEM37 | 1554485_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM37 | 1554485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.012
p=0.0336
p=0.5366
p=0.2477
p=0.1269
p=0.944
p=0.2328
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2